

A V Sharmili

SUMMARY

This internship enhanced my knowledge of Data Science, Machine Learning, Python programming, and industrial data analytics while providing valuable practical experience in solving a real-world industrial problem.

EDUCATION

Mangayarkarasi College of Engineering

2026

EXPERIENCE

Internship Project

2026

Universal Consulting Technologies (UCT)

Quality Prediction in a Mining Process Using Machine Learning

- Developed a Machine Learning model to predict % Silica Concentrate in an iron ore flotation plant before laboratory analysis.
- Selected and trained an XGBoost Regressor model using operational process parameters.
- Performed data preprocessing including handling missing values, date conversion, feature selection, and train-test split (80/20).
- Evaluated model performance using MAE, RMSE, and R^2 Score.
- Generated visualizations: Actual vs Predicted Silica Concentration, Prediction Error Distribution, and Feature Importance Chart.
- Implemented an early warning system with a 3.0% silica threshold for proactive operational decisions.

PROJECTS

Quality Prediction in a Mining Process | Machine Learning-based prediction of silica concentrate in an iron ore flotation plant using XGBoost Regressor.

- Target variable: % Silica Concentrate.
- Model parameters: Number of Estimators 200, Learning Rate 0.05, Maximum Depth 6, Subsample 0.8, Column Sample by Tree 0.8, Random State 42.
- Workflow included dataset collection, cleaning, preprocessing, feature selection, training, prediction, evaluation, visualization, and early warning generation.

TECHNICAL SKILLS

Python

Machine Learning

Data Science

Data Cleaning

Data Preprocessing

XGBoost Regression

Feature Selection

Model Evaluation

Data Visualization

GitHub Version Control

Problem Solving

Analytical Thinking

Technical Documentation
Time Management
Industrial Data Analysis
Pandas
NumPy
Matplotlib
Scikit-learn
XGBoost
Google Colab
Visual Studio Code

INTERNSHIP OBJECTIVES

Learn Data Science concepts; Understand Probability and Statistics; Study Machine Learning fundamentals; Apply Machine Learning techniques to industrial datasets; Develop practical programming skills using Python; Gain experience using GitHub for project management; Improve problem-solving and analytical thinking.

CHALLENGES FACED

Handling missing values; Data preprocessing; Converting data into numerical format; Selecting relevant features; Understanding model performance metrics.

FUTURE ENHANCEMENTS

Hyperparameter optimization; Real-time prediction using sensor data; Web application deployment using Streamlit; Integration with Industrial IoT systems; Cloud-based monitoring dashboards.

REFERENCES

Python Documentation; Pandas Documentation; NumPy Documentation; Matplotlib Documentation; Scikit-learn Documentation; XGBoost Documentation; Mining Process Flotation Plant Dataset; Universal Consulting Technologies (UCT) Internship Guidelines.